

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Integrated Digital Climate Advisory Services for 6 million Farmers through PPP in Kenya

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 1 · Close the Climate Adaptation Gap through Scalable Climate Services and Smart Water Management

Flagship 1.1 · Digital Climate Services for Resilient Farm & Enterprise Management

Version 1.0 · June 2026

Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Kenya's smallholder farmers, the backbone of the nation's food system, face increasing climate volatility, yet access to actionable, localized advice remains critically limited. While Kenya possesses world-class institutional assets, including the Kenya Meteorological Service Authority (KMSA) for forecasts and the Kenya Agricultural and Livestock Research Organization (KALRO) for agronomic expertise, these systems remain fragmented. Forecasts are not consistently translated into value-chain specific advisories, and traditional extension reaches fewer than 20% of farmers. This solution track addresses this gap by scaling an integrated, end-to-end digital climate advisory service (DCAS) for seven million farmers by 2028.

The solution is a bundled innovation that connects KMSA forecasts, KALRO's advisory platform (KAOP), and Kenya's national farmer registry into a unified digital public infrastructure. This integration enables the automated translation of weather data into localized, multilingual advisories for crops, livestock, and rangelands, delivered via SMS, USSD, IVR, and a network of 3,000+ extension agents. The approach moves beyond pilot projects to institutionalize a government-owned, financially sustainable model, leveraging a public-private partnership (PPP) anchored in Kenya's national digital agriculture strategy. Core implementing partners include KALRO, KMSA, the Ministry of Agriculture and Livestock Development (MOALD), and key development programs like the World Bank's NAVCDP and FSRP. This track is supported by a strong evidence base from Kenya and beyond, demonstrating that digital advisories can deliver significant yield gains (up to 12%) at a fraction of the cost of traditional extension (less than US\$1 per farmer).

2. Innovation Bundle and Scaling Ambition

What is Being Scaled

- **Integrated Digital Public Infrastructure (DPI):** The technical and institutional integration of KMSA's forecasts, KALRO's KAOP platform, and the national farmer registry into a single, interoperable advisory pipeline.
- **AI-Enabled, Multi-Channel Advisory System:** A system that uses AI-assisted translation and personalization to generate localized advisories for 20 crops and livestock systems, delivered via SMS, USSD, and Interactive Voice Response (IVR) in multiple local languages.
- **Hybrid Public-Private Delivery Model:** A scaled network of 3,000+ extension agents and agripreneurs, reinforced by private sector partners (e.g., telecoms, input suppliers) and NGO implementers (e.g., Digital Green, Kuza) for last-mile reach.
- **Climate-Smart Decision-Support Modules:** Content packages for specific value chains, including crop calendars, livestock thermal stress alerts, and rangeland management tools.

Evidence of Impact

- A meta-analysis in Sub-Saharan Africa shows that digital information interventions yield average productivity gains of 6–9% and increase fertilizer adoption (Beach et al., 2025).
- In India, scaled digital advisory programs generated ~4% average yield increases, demonstrating modest but consistent effects at scale (Fabregas et al., 2019).
- In Kenya, KALRO/KMSA SMS-based agro-weather advisories reached over a million farmers, with 50% of surveyed farmers willing to pay ~\$0.58 per month (Kamau et al., 2024).
- The TomorrowNow Digital Weather Advisory Service in Kenya showed farmers achieved a 12% yield increase at a cost of \$0.56 per farmer, with 88% reporting the forecasts as accurate and useful (TomorrowNow, 2024).

Scaling Targets

Farmers actively reached	1 million (in 10 counties)	6 million (approx. 70% of national registry; 50% women, 40% youth)
Area under improved practices	0.5 million hectares	2 million hectares managed under climate-smart practices
Jobs supported	1,000 extension agents/agripreneurs trained	3,000+ extension agents and agripreneurs activated
Investment leveraged	Leverage ~\$5M from AICCRA III and other programs	Secure 30% of operational costs from domestic public/private sources

3. Scaling Pathway and Theory of Change

Pathway Strategy

The scaling pathway is a hybrid public-private partnership (PPP) model designed for institutionalization within Kenya's national extension system. The core strategy is to act as a "systems integrator," connecting existing public assets with private-sector delivery to create a sustainable, national-scale service.

The strategy unfolds in three parallel streams:

- **National Rollout via Public Infrastructure:** The foundation is KALRO's KAOP platform, which integrates KMSA forecasts and connects to the national farmer's registry. S4I funding will support the technical upgrades (APIs, AI tools) needed to automate and personalize advisory generation at scale.
- **County-Level Embedding:** The track will work directly with county governments to embed DCAS into the workplans of major donor programs like the World Bank's NAVCDP and FSRP. This ensures the service is integrated with the government's agripreneur network and local extension agents, securing local ownership and operational budgets.
- **Private and Development Leverage:** Strategic partnerships with telecom operators, agribusinesses (e.g., Yara), and NGOs (e.g., Digital Green, Kuza, AGRA) will extend reach and explore bundled value-added services. This creates a pathway for private co-financing (e.g., sponsored SMS campaigns) and ensures the system can adapt to diverse user needs.

Theory of Change

If Kenya's fragmented climate and agriculture assets (forecasts, agronomy, farmer data) are integrated into a unified digital public infrastructure, **and** this system is used to deliver timely, localized, and multilingual advisories through a mix of public and private last-mile channels, **then** farmers and intermediaries will adopt more proactive climate risk management practices. This will lead to **increased productivity and resilience for millions of farmers**, and the institutionalization of the service within Kenya's national extension strategy will ensure **long-term sustainability and systemic change**.

Key Activities Funded by S4I

- **Technical Integration:** Develop automated APIs between KMSA and KAOP; deploy AI-enabled content assembly for personalized, multilingual advisories.
- **Partnership and Governance:** Convene and facilitate a multi-stakeholder coalition (KALRO, KMSA, MOALD, county governments, NAVCDP/FSRP) to align strategies and formalize agreements; develop a roadmap for embedding DCAS into the National Agricultural Extension Strategy.
- **Inclusive Delivery and QA:** Co-design and test inclusive delivery models (IVR, HCD with women/youth); establish a human-in-the-loop quality assurance process for all AI-generated advisories.
- **Monitoring, Learning and Evaluation:** Implement an A/B testing framework with PxD; deploy real-time monitoring via USSD polls and call data; conduct annual outcome surveys to track adoption and behavioral change.

4. Partners and Roles

Partner Type		Specific Partners and Quantities	Role in Scaling Pathway
Institutional Partners	Scaling	Ministry of Agriculture (MOALD), KALRO, KMSA, Kenya Agricultural Data Information Centre (KADIC), 20+ County Governments	Ownership and Institutionalization: Provide core infrastructure (KAOP, forecasts, registry), set policy, integrate into county budgets and national extension strategies, and ensure long-term government stewardship.
	Demand and Last-Mile Partners	NAVCDP, FSRP (World Bank); AICCRA III, IFAD PPP; Digital Green, Kuza, AGRA (VBAA network), WFP-FtMA; 3,000+ extension agents/agripreneurs	Reach and Delivery: Deliver advisories to farmers, provide feedback loops, bundle services, and ensure the system is responsive to user needs. They represent the primary channel for achieving scale.
Technical and Research Partners		ILRI, IFPRI, AIM4Scale, UCSB/FEWS NET, Strathmore SAFIC, PxD	Innovation and Learning: Provide scientific rigor in livestock advisories (ILRI), AI (IFPRI), forecast downscaling (UCSB), capacity building (SAFIC), and evidence-based learning (PxD).

Funding Model: S4I catalytic funding provides the critical investments for **technical integration, partnership coordination, and learning systems (CAPEX and core coordination)**, which in turn unlocks and aligns **~\$500M+ in existing bilateral investments (NAVCDP/FSRP) and future donor funding (AICCRA III, IFAD)** that finance the core operational costs of outreach, training, and infrastructure support.

5. Enabling Environment and Risks

Opportunities

- **Strong Institutional Foundation:** Kenya has a national farmer registry, a functional met agency (KMSA), and a respected agricultural research organization (KALRO), providing the core public assets needed for an integrated system.

- **Major Aligned Investments:** The presence of large-scale programs like the World Bank's NAVCDP/FSRP (~\$500M) and the upcoming AICCRA III and IFAD PPP provides a massive pool of operational funding to co-finance the rollout.
- **Demonstrated Government Demand:** Alliance MOUs with KMSA and KALRO; active requests from KMSA on business models with PPP focus and KALRO for technical support (e.g., IVR automation) confirm institutional demand.
- **Proven Farmer Demand:** Evidence shows >80% of farmers find SMS advisories useful and ~50% are willing to pay, creating a strong pull from end-users and potential for private sector engagement.

Challenges and Mitigation

- **Political Economy and Institutional Silos:** The primary risk is that KALRO, KMSA, and other agencies may resist integration due to competing mandates or territoriality. **Mitigation:** The track's primary focus is on **convening and convincing**. Activities will prioritize building a coalition of champions, formalizing partnerships through MOUs, and leveraging the influence of major donors (World Bank, IFAD) to align incentives around a shared national goal.
- **Financial Sustainability Beyond Donors:** Operational costs (OPEX) for staffing, cloud services, and hardware refresh could become unfunded once donor projects end. **Mitigation:** The track explicitly targets **30% domestic financing by 2027** and full institutionalization in the National Extension Strategy by 2028. This requires a dedicated workplan to embed costs into MOALD and county budgets, and to pilot tiered pricing/freemium models with private partners.
- **Digital Divide and Social Inclusion:** Older farmers, women, and pastoralists may have lower digital literacy and access. **Mitigation:** The pathway incorporates inclusive design from the outset: IVR in local dialects, community radio integration, group listening sessions, women-only trainings, and a youth "digital advisor" program. Human-centered design (HCD) cycles will specifically target these groups.
- **Quality and Trust in AI-Generated Advisories:** Poorly designed advisories could lead to maladaptation and loss of trust. **Mitigation:** A strict **human-in-the-loop validation process** will be used. All AI-generated content will be reviewed by KALRO/KMSA experts against national climate-smart agriculture guidelines before dissemination.

6. Learning and Adaptation

Monitoring Focus

- **Process and Uptake:** Real-time monitoring of advisory delivery (SMS, IVR), USSD poll responses, and call center data via the KAOP platform to track reach and user engagement.
- **Behavior Change:** A/B testing frameworks co-developed with PxD to evaluate which message formats, channels, and content drive the highest rates of adoption and practice change among farmers.
- **System Outcomes:** Annual outcome surveys to measure yield, income, and resilience impacts, with data disaggregated by gender and age to track equity goals. Partner feedback surveys assess the effectiveness of the coordination model.

Adaptive Management

- **Quarterly Solution Track Check-ins:** Regular meetings with the S4I team to review progress against milestones, analyze monitoring data, and adapt workplans based on emerging evidence.

- **Bi-Annual Joint Learning Workshops:** Convene all core partners (CGIAR, KALRO, KMSA, counties, donors) to review findings from A/B tests and surveys, share lessons, and collaboratively adjust the scaling strategy for the following season.
- **Dedicated Scaling Advisor Support:** The S4I team will provide a dedicated scaling advisor to help the lead scientist navigate partnership challenges, refine the sustainability model, and ensure activities remain focused on scaling outcomes rather than upstream research.

7. Next Steps and Timeline (2026)

Quarter	Key Activities
Q1 (Jan-Mar)	• Finalize selection of 10 priority counties in consultation with KALRO/KMSA • Formalize pending Alliance MOU/MOA with key partners • Co-design detailed 2026 workplan with S4I and AIM4Scale teams
Q2 (Apr-Jun)	• Begin technical integration: Design the workflow to facilitate data flow between KMSA and KAOP based on consultation with the respective technical teams • Initiate human centric design (HCD) scoping with target extension agents and farmer groups (women, youth, pastoralists) • Training material development for first cohort of extension agents/agripreneurs • Conduct stakeholder workshop to align DCAS integration with NAVCDP/FSRP county workplans
Q3 (Jul-Sep)	• Design push-and-pull advisory services (SMS, USSD, IVR) in 5 initial counties • Begin A/B testing of message formats with PxD (or other experts) • Initiate work on the roadmap for embedding DCAS into the National Agricultural Extension Strategy • Training of 1150 agents/agripreneurs (this can also happen in Q4)
Q4 (Oct-Dec)	• Scale rollout of advisory services to all 10 target counties, reaching 1M farmers • Complete first round of HCD feedback cycles and incorporate learnings into v2.0 of the advisory system • Present annual outcome report and scaling progress at a joint learning workshop

Communications and Knowledge: Key products will include a project launch brief, quarterly blog posts highlighting lessons from HCD and A/B testing, and a high-impact annual report featuring farmer success stories and quantitative progress against targets. Channels will include the CGIAR website, partner newsletters, and donor-facing presentations.

8. Contact

- **Aniruddha Ghosh (Alliance of Bioversity International and CIAT)** | a.ghosh@cgiar.org
- **Evan Girvetz (Alliance of Bioversity International and CIAT)** | e.girvetz@cgiar.org